AI-compute supply chain

Chart pack based on analysis of data from UN Comtrade + TDM + Trade Atlas, from which we construct a bilateral monthly panel of HS6 trade flows capturing the global AI-compute supply chain (60 HS6 codes, 2017-01 to 2026-04).

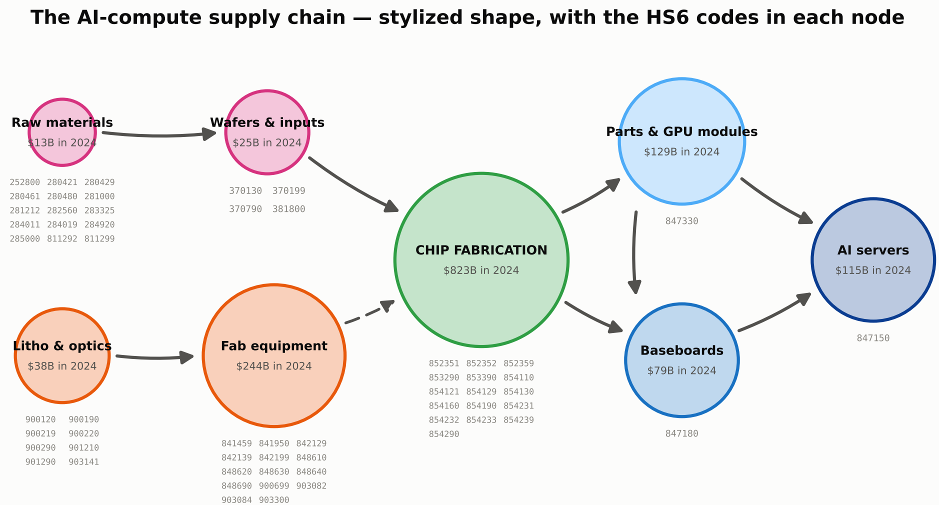
2026-08-20

1. The supply chain in 2024

2024 annual trade flows from the Trade Atlas. We record all relevant HS6 codes, so its coverage extends beyond solely AI compute. China and Hong Kong are counted as one trade bloc (CHK) throughout.

Topography of the supply chain

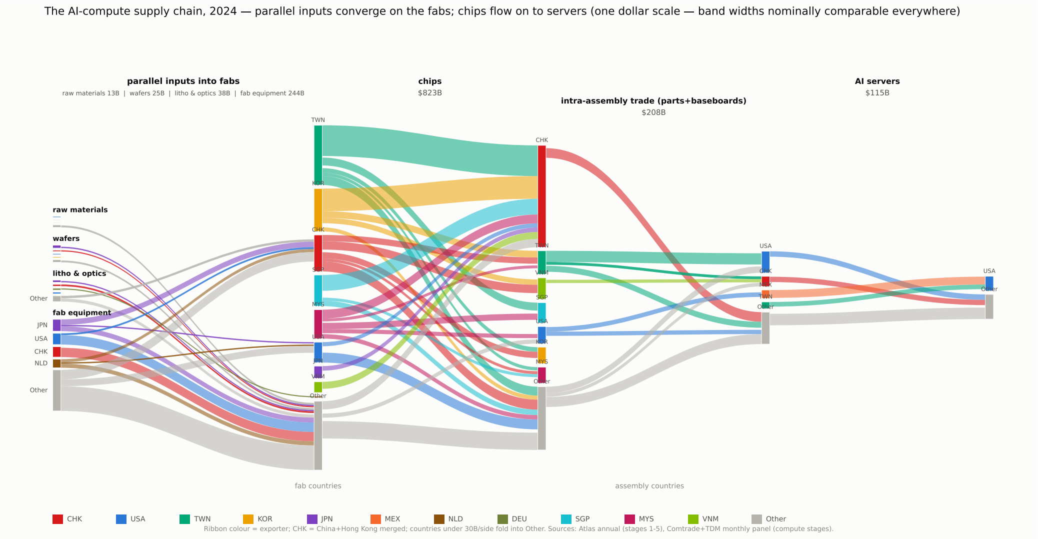
Two upstream inputs feed chip fabrication: capital equipment in orange is used to build manufacturing facilities, raw materials and wafers are consumed per chip. Green represents the actual chip fabrication, which feeds downstream assembly toward AI servers in blue. Circle size in proportion with the log of 2024 USD total trade value.



Two input branches, one output line. Materials (pink) -- silicon, chemicals, wafers -- are used up with every chip made (solid arrows). Equipment (orange) -- lithography and other fab tools -- is investment in capacity: in Taiwan and Korea, equipment imports rise six to twelve months before chip exports (dashed arrow). Korean memory and Taiwanese chips are packaged together, usually in Taiwan. Chip design earns as a service and crosses no border. Circle diameters grow with the log of 2024 world trade. Blue marks the three codes of our monthly panel (the Fed AI-compute basket). A few niche inputs (lasers, vacuum pumps and valves, chip substrates) fall outside the 60 codes. Codes: OECD semiconductor mapping + Fed basket (docs/data.md, section 1).

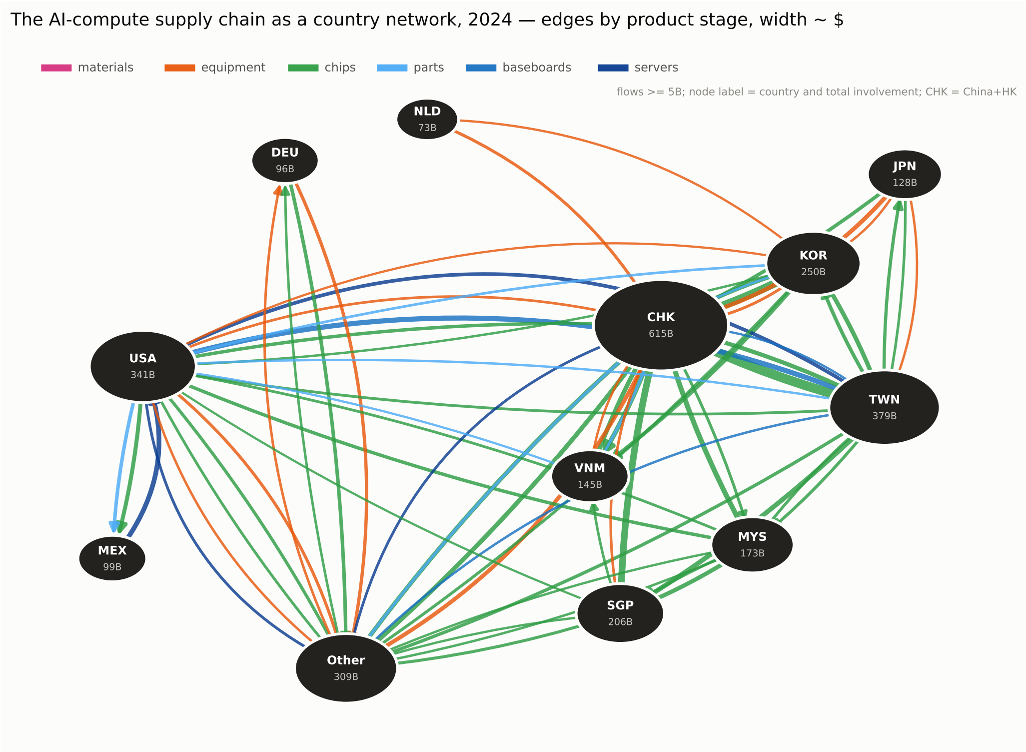
The chain in bilateral trade flows

Width of band = 2024 USD trade. Small flows are grouped into other (grey).



The chain as a graph network

Countries are represented as nodes and directional flows as edges, colored by stage.



2. The chain in stages

2024 annual trade flows from the Trade Atlas. We estimate a matrix factor model to summarize trade flows by exporting and importing hubs.

The matrix factor model

Per stage, the bilateral export matrix, X, has i-jth entry giving the export value from country i to country j in 2024. The Matrix Factor Model (MFM) for X is given by

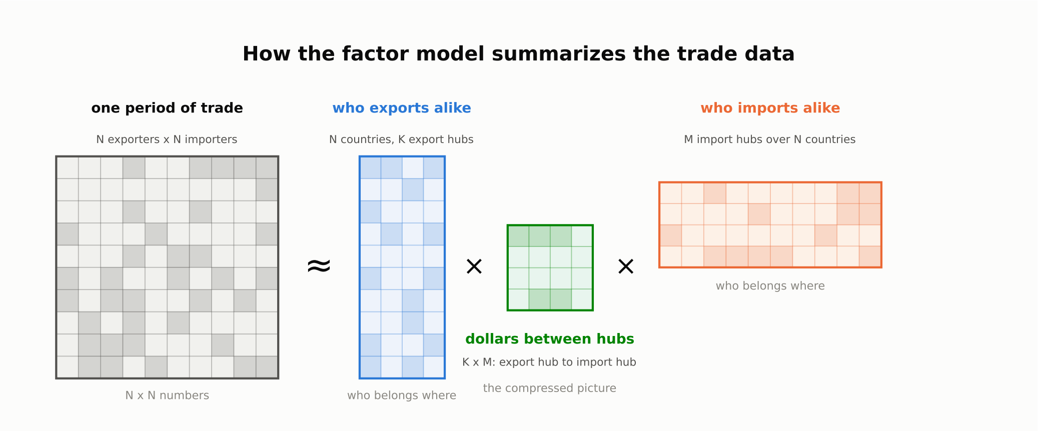
$$X = R\,F\,C' + e,$$

for N countries, K export hubs indexed by k, and M import hubs indexed by m:

- R $(N \times K)$ are the **export loadings**: the weight of each country in each export hub.
- C $(N \times M)$ are the **import loadings**: the weight of each country in each import hub.
- F $(K \times M)$ are the **matrix factors**: F_{km} is the dollar flow from export hub k to import hub m.
- e $(N \times N)$ are the idiosyncratic errors.

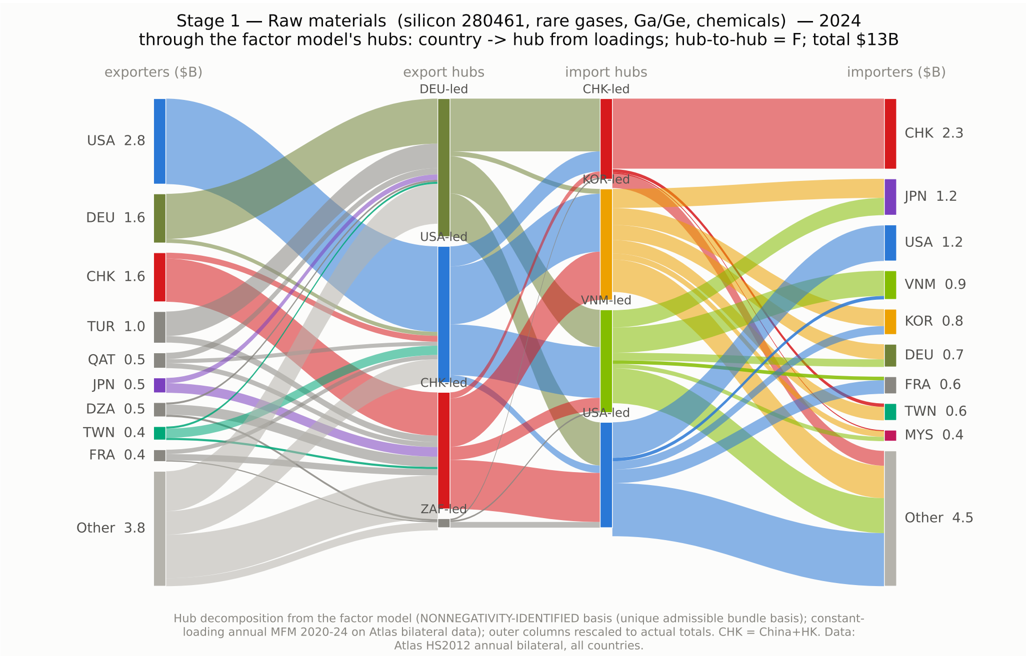
The same model as a picture

Similar to a PCA model, which summarizes co-movement in a vector as loadings times a low-dimensional vector of factors, the MFM does the same for a matrix: it summarizes X by a low-dimensional matrix of factors with two sets of loadings.



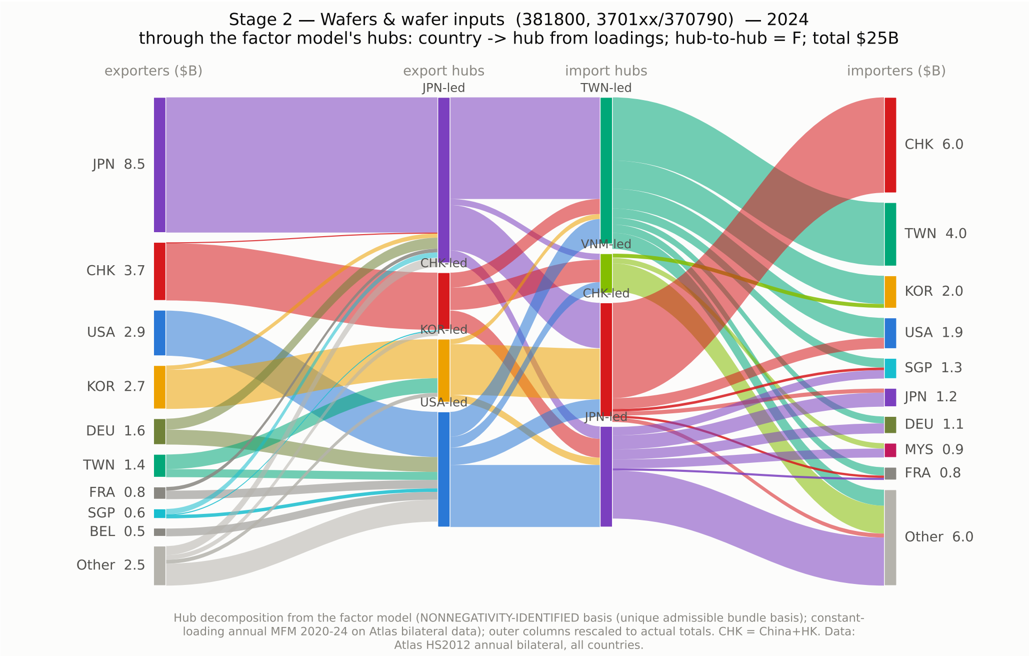
Stage 1 – raw materials

Left to right: exporters to export hubs to import hubs to importers. Raw materials trade is relatively small (13B) and diffused across countries.



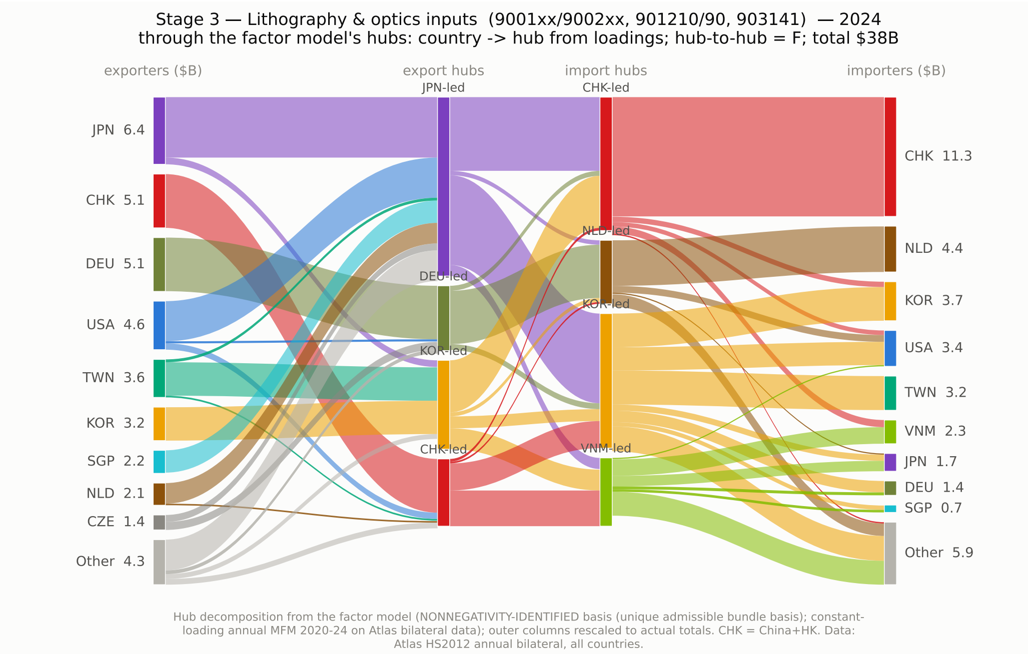
Stage 2 – wafers and wafer inputs

Japan is the largest exporter, about a third of the 25B world total.



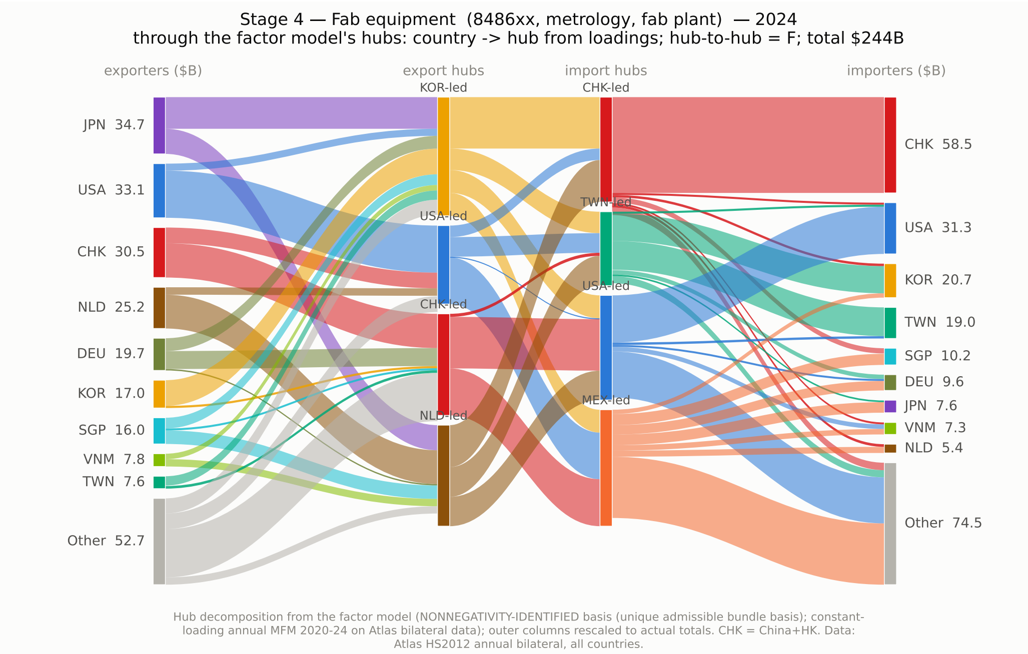
Stage 3 – lithography and optics inputs

Lenses, optical elements, electron microscopes and wafer inspection tools. No dominant supplier: Japan, the China bloc, Germany and the United States each export 5 to 6B.



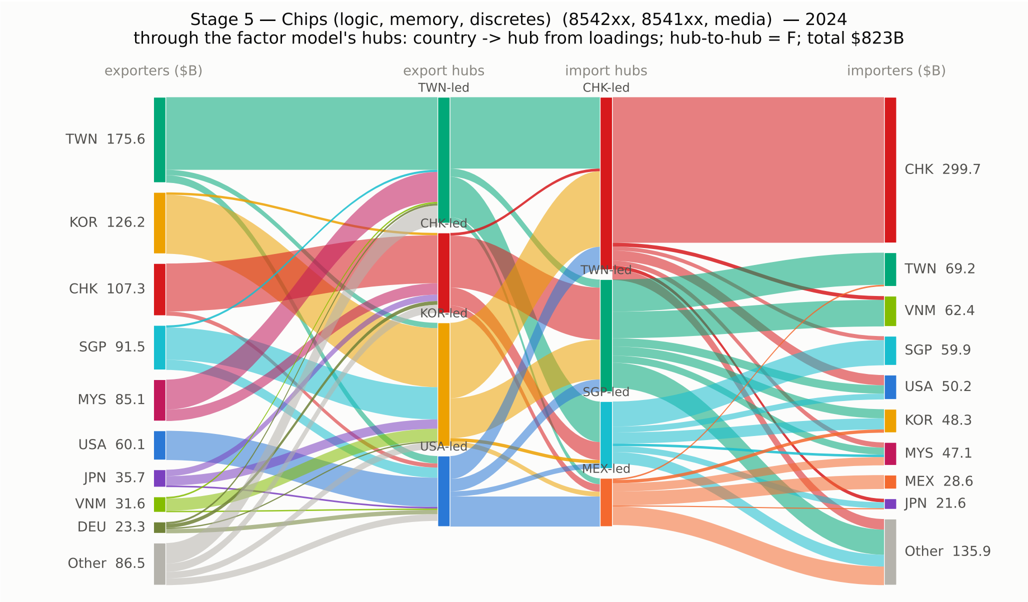
Stage 4 – fab equipment

Machines, not materials: countries buy these to build capacity, and their imports rise 6-12 months before chip exports.



Stage 5 – chips

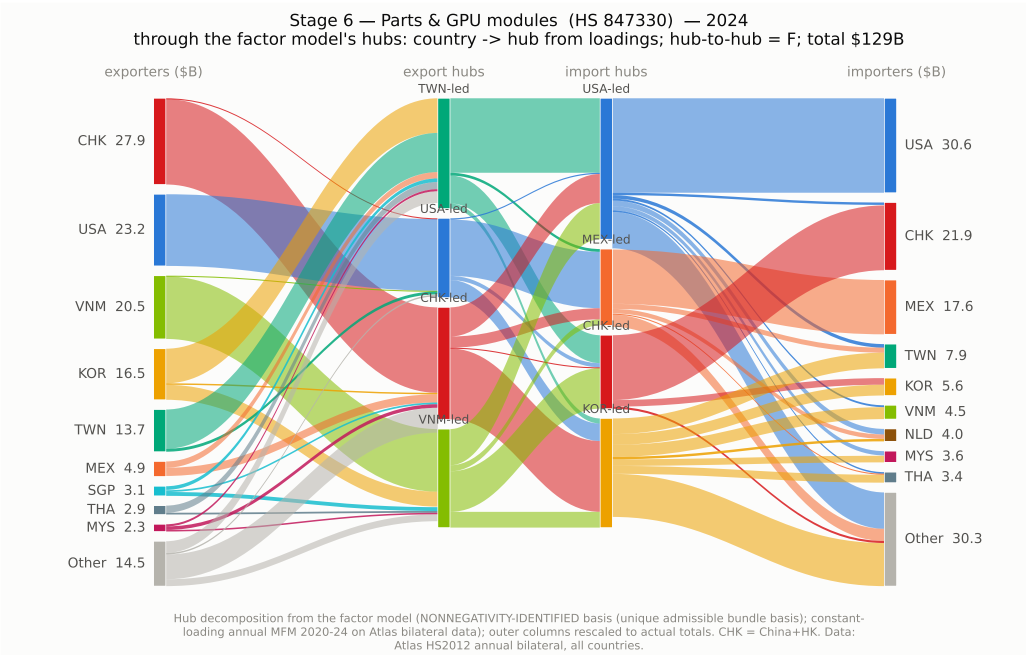
The biggest stage, and the whole electronics industry – phones, cars and PCs, not just AI. Most chips flow to Asian assembly; the China sphere absorbs about 43%.



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 on Atlas bilateral data); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral, all countries.

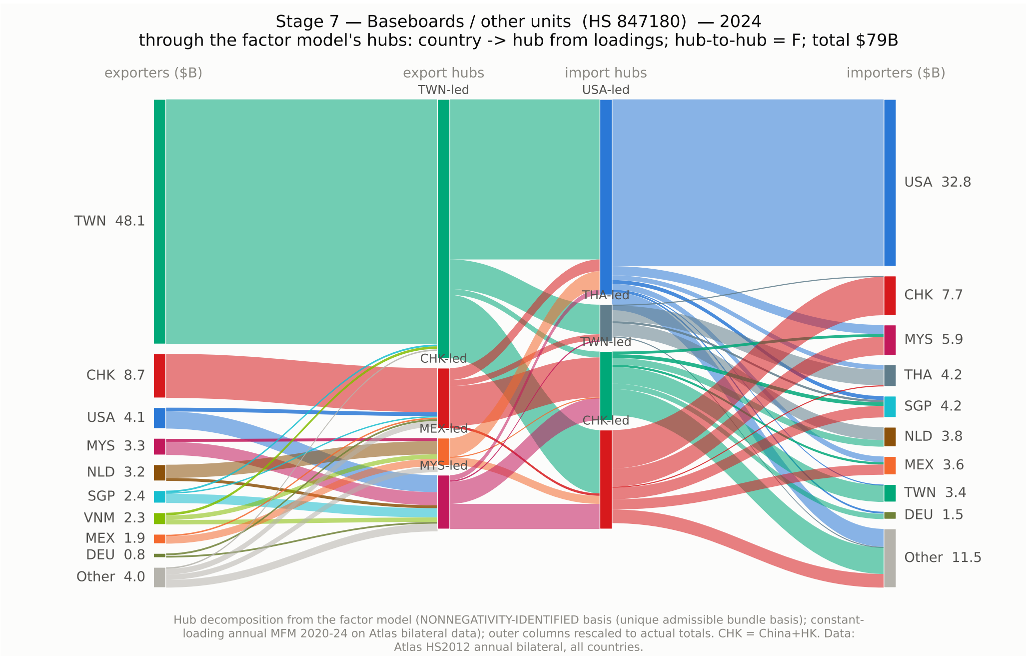
Stage 6 – parts and GPU modules (847330)

The China bloc exports the most by value (28B in 2024), ahead of the United States and Vietnam; the United States is the largest buyer (31B), then the China bloc and Mexico.



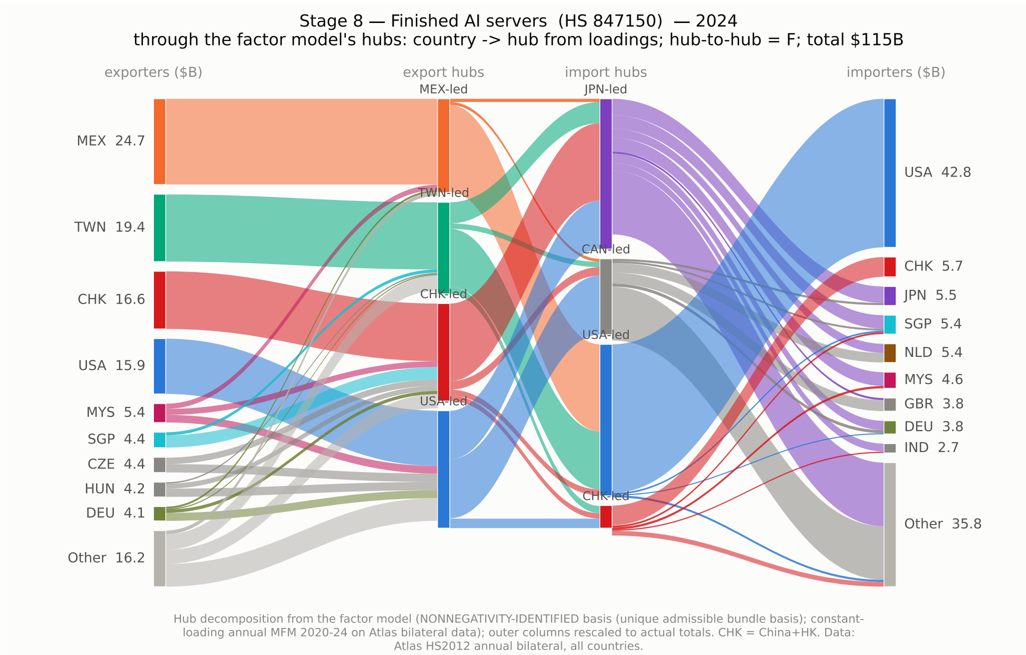
Stage 7 – baseboards (847180)

The most concentrated stage: Taiwan exports 48B of the 79B total in 2024, five times the next exporter, and the United States buys 42 percent of world imports.



Stage 8 – AI servers (847150)

Mexico (25B) and Taiwan (19B) assemble and ship; the United States buys 43B, 37 percent of world imports in this code.

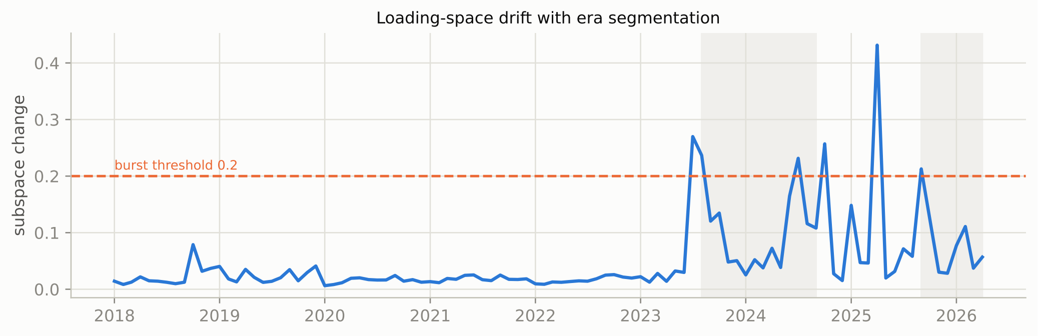


3. The time-varying MFM – AI-compute only

This section looks at the three downstream AI-compute HS6 codes jointly (parts/GPU modules, baseboards, AI servers), and estimates a time-varying MFM on monthly trade from 2017.

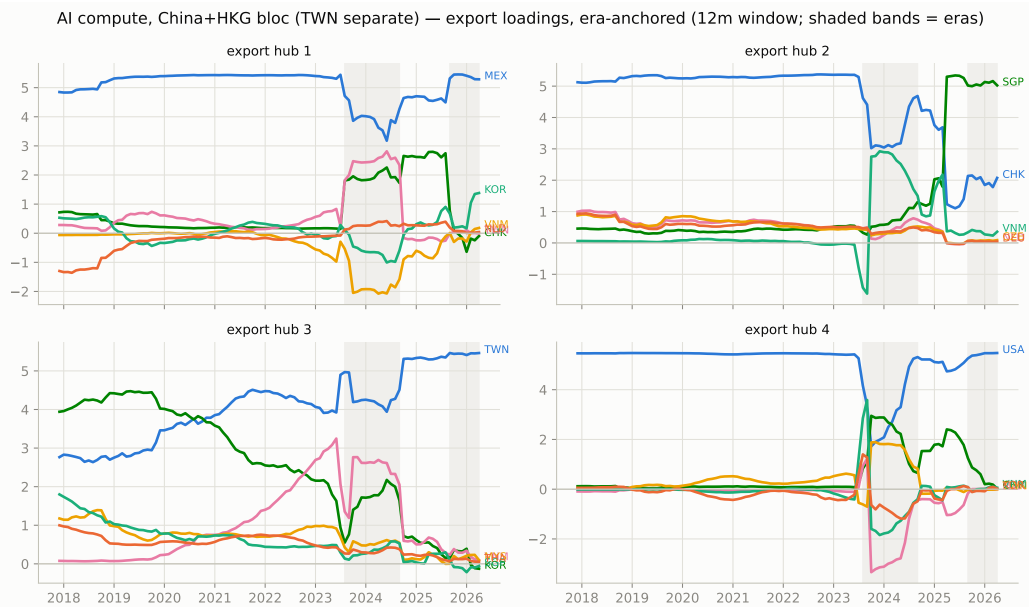
When the network changed shape

The line measures how much the time-varying export groupings moved month-to-month. We see stability until Aug 2023 (AI demand), Oct 2024 (memory export controls were signaled), and Sep 2025, where the hubs simplify to roughly one country each; no policy event is established for that last break.



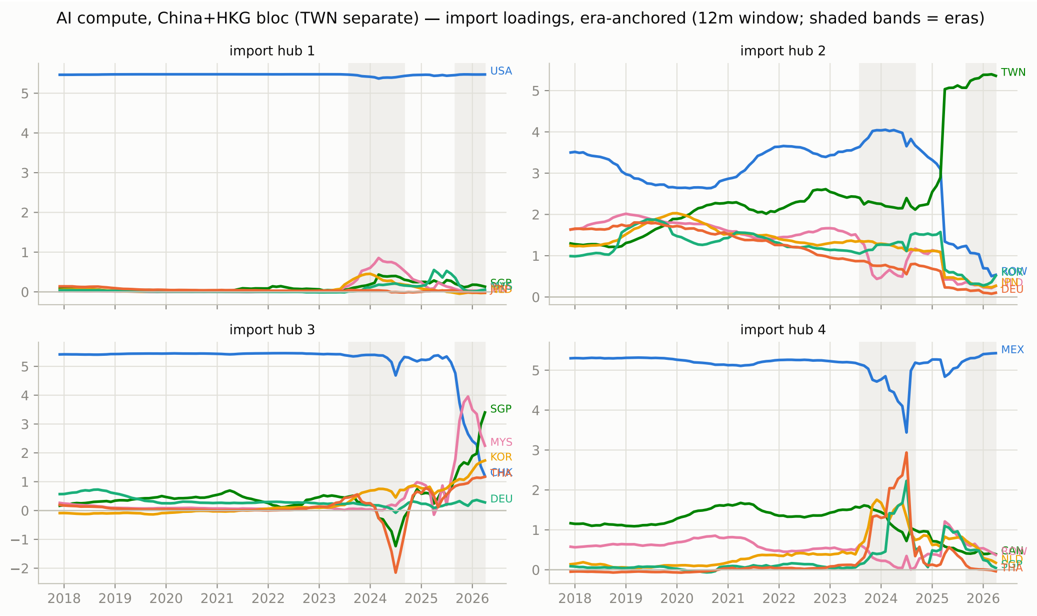
Who is in each export group

At Oct 2024 the China group loses its separate identity and a Singapore-led group appears.



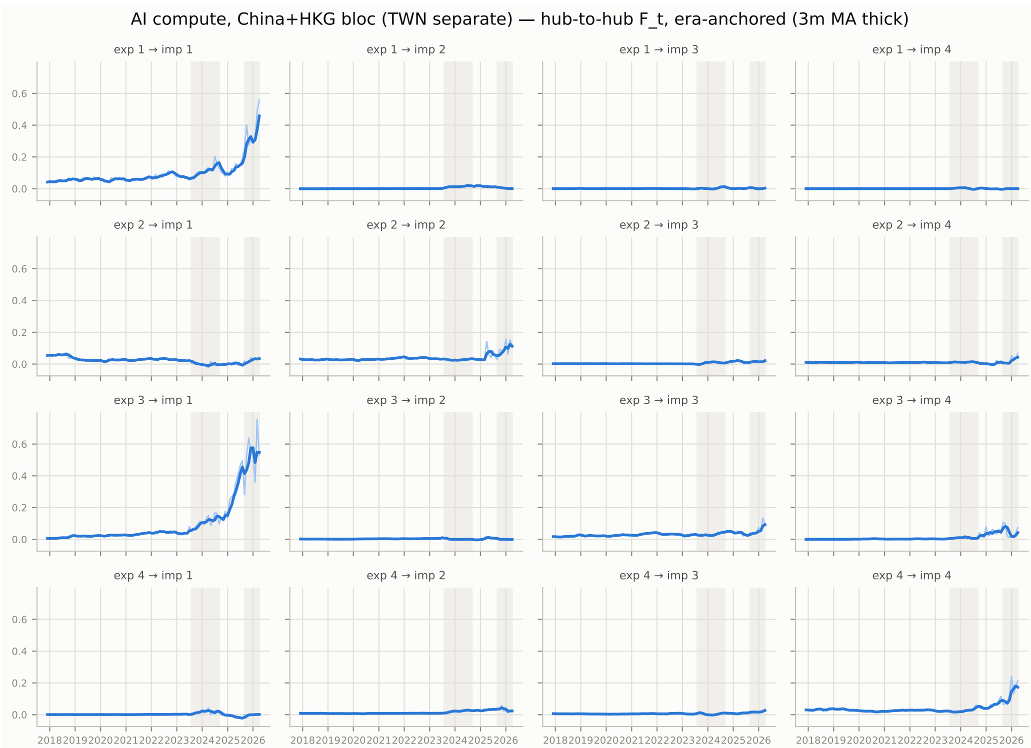
Who is in each import group

The import side is steady: a US group, a Mexico group, a Taiwan group, and the rest of the world.



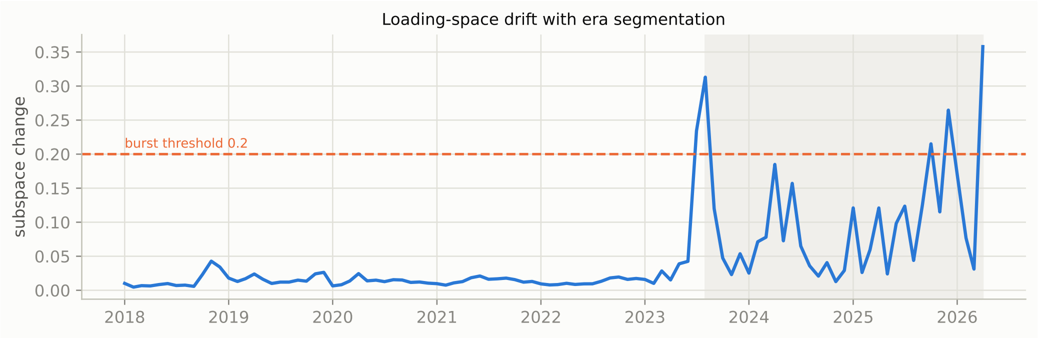
Dollars between groups, month by month

Three of the sixteen channels carry the AI boom: Taiwan to the US, and both legs of the US-Mexico assembly loop. The rest are flat.



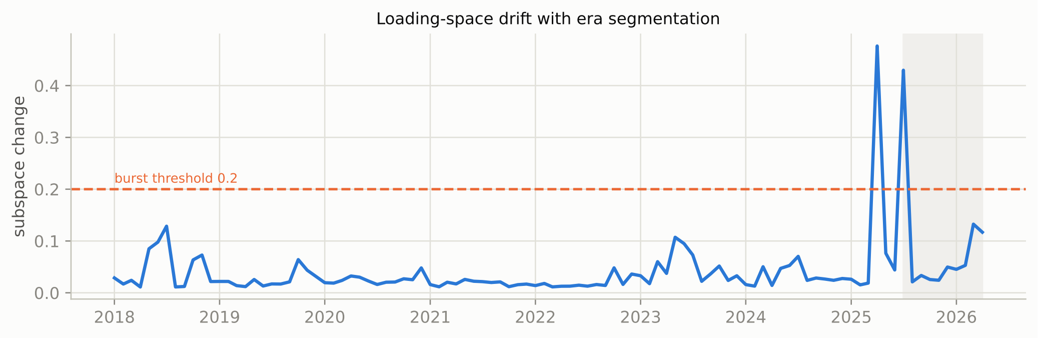
Same, with every country separate

Without merging China and Hong Kong there is one break in the whole sample: Aug 2023. Demand changed the country-level structure; policy did not.



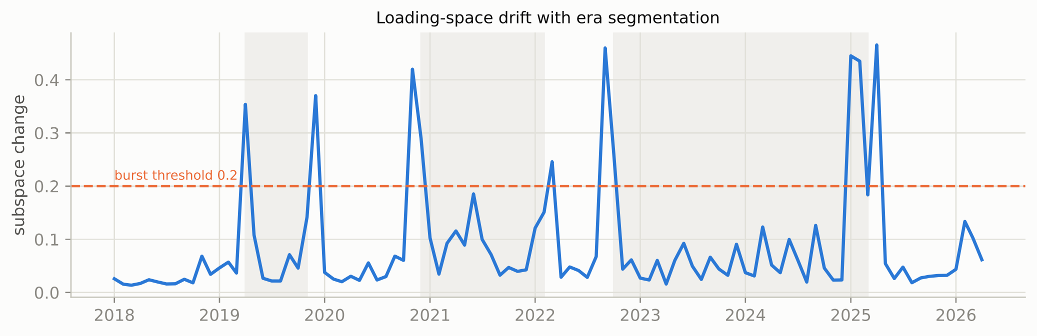
Parts only (847330)

91 months without a break – the stable base layer of the network.



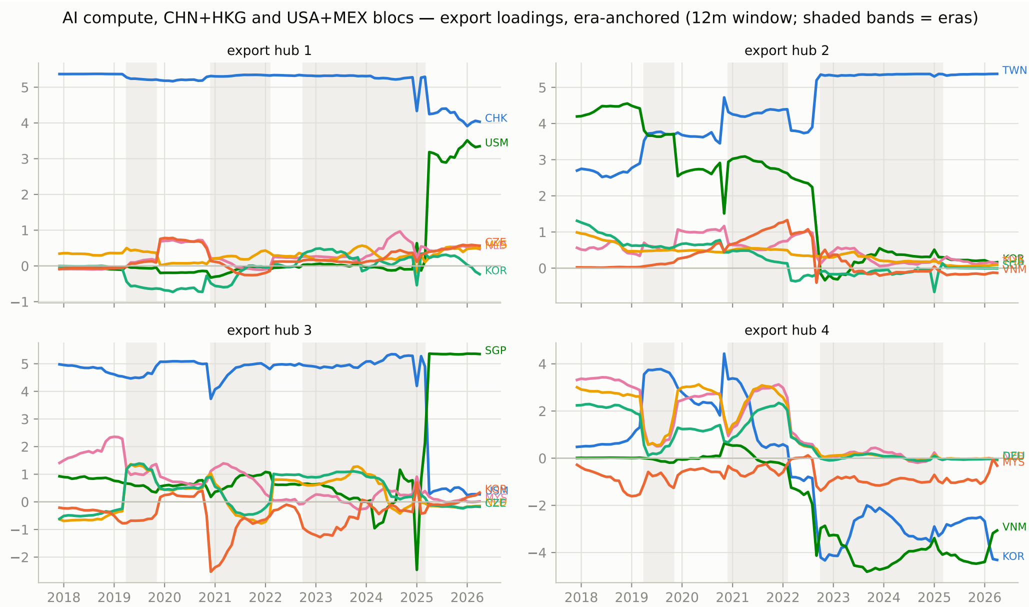
With both blocs merged

Merging China+Hong Kong and US+Mexico hides churn inside each bloc and isolates what happens between them. The breaks line up with policy: the 2019 tariff war, the Oct 2022 export controls, the Apr 2025 tariffs.



The tariff signature

After Apr 2025 the two blocs move together on a single factor – seen at no other break.

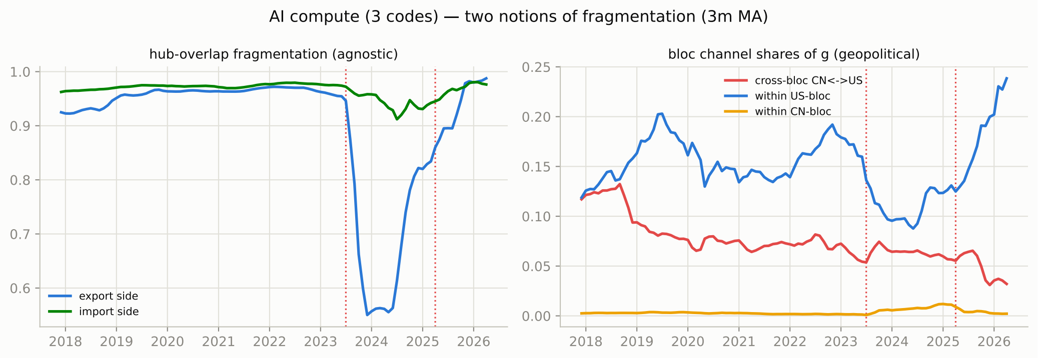


4. Concentration & fragmentation – AI-compute codes only

Monthly measures built on top of the model, for the three blue codes only. Dotted lines mark Jul 2023 and Apr 2025. Interpretation: results/network_stats/findings.md.

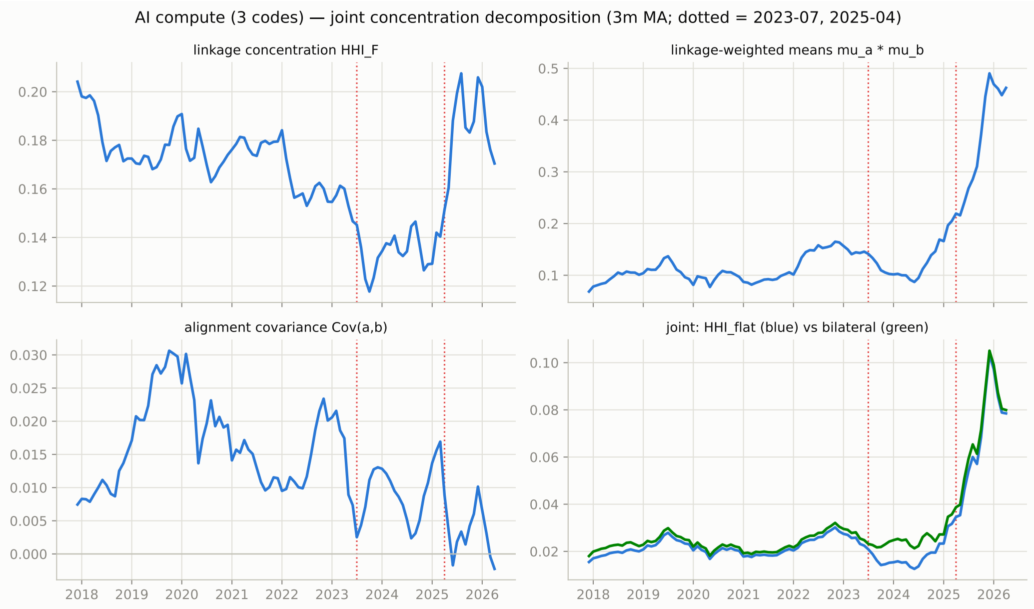
Two meanings of fragmentation

Left: do the model's groups overlap (a technical check, little economic signal). Right: the share of trade crossing between the China and US blocs – down about 70% since 2017, in two steps (2019, then 2023 on), with no recovery between.



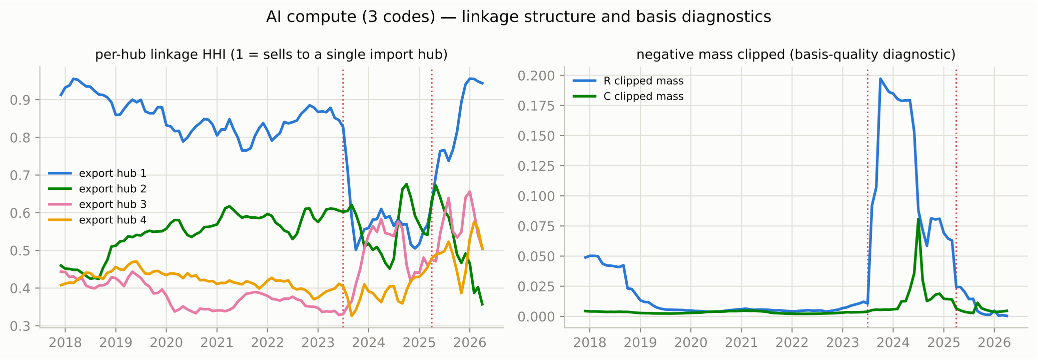
Concentration, taken apart

Overall concentration splits exactly into three parts. The interesting one asks: do concentrated sellers ship to concentrated buyers? Mildly yes, peaking around 2019, fading through the AI era.



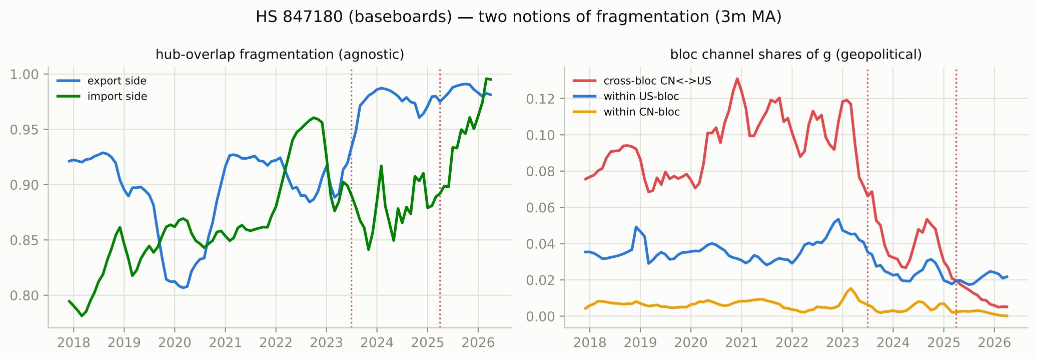
One buyer or many?

Left: does each export group sell into one import group or several. Right: a data-quality check, elevated during the 2023-24 transition.



Baseboards: the sharpest split

The cross-bloc share collapses about 20x from its 2021 peak – the code most exposed to export controls.



Parts: still shared

The cross-bloc share halves but stays the highest of the three codes: the parts trade remains the layer both blocs still share.